HDOC DAY 68

Decimal to Binary Using Recursion

Aim Convert a nonnegative decimal integer into its binary representation using a recursive function.

Algorithm

1. Read a nonnegative decimal integer.

2. If n is greater than 1, recursively process n/2 first.

3. Print n%2 while returning from recursion.

4. For input 0, print 0 directly.

C Program

```
#include <stdio.h>

void print_binary(unsigned int n)
{
    if (n > 1)
        print_binary(n / 2);
    printf("%u", n % 2);
}

int main(void)
{
    unsigned int n;
    printf("Enter decimal number: ");
    if (scanf("%u", &n) != 1) {
        printf("Invalid input.\n"); return 1;
    }
    printf("Binary representation: ");
    print_binary(n);
    printf("\n");
    return 0;
}
```

Test Cases

Input	Expected result
0	0
10	1010
25	11001

Compilation and Execution (Mac Terminal)

```
clang -std=c99 -Wall -Wextra day68.c -o day68
./day68
```

Sample Output

```
Enter decimal number: 25
Binary representation: 11001
```

Important Note

Recursing before printing produces bits from most significant to least significant. The program is intended for nonnegative integers.

Result

The program was implemented using the required technique and produces the expected results for the given test cases.